

Industrial & Systems Engineer (Ph.D.) | Simulation Solution Consultant | AI  
Solution Architect

# JEHUN LEE

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**Scholar:** <https://scholar.google.com/citations?user=C7ekyjEAAAAJ>



## Professional Profile

In the AI era, the most important question is not which model to use, but what problem is worth solving — defined through human will and necessity. Algorithms don't choose direction; people do.

My direction is to be a keyman: someone who sees the entire system, maps stakeholders, sets priorities, and deeply understands the problems that need solving so direction can be defined. The growth that lasts is the growth where the organization grows together — not where individuals optimize alone.

Systemic excellence, to me, means hiring AI agents — treating them as employees who execute autonomously, with humans setting direction and confirming key decisions. Chores, basic reviews, missing numbers move into agent pipelines, freeing human attention for the work only humans can do: judgment, strategy, meaning.

## Education

|                   |                                                      |                                                            |
|-------------------|------------------------------------------------------|------------------------------------------------------------|
| 2021.03 - 2025.02 | <b>Ph.D. in Industrial &amp; Systems Engineering</b> | Korea Advanced Institute of Science and Technology (KAIST) |
| 2016.09 - 2021.02 | <b>M.S. in Industrial Engineering</b>                | Sungkyunkwan University (SKKU)                             |
| 2013.03 - 2016.08 | <b>B.S. in Systems Management Engineering</b>        | Sungkyunkwan University (SKKU)                             |
| 2011.03 - 2013.02 | <b>High School in Math &amp; Science Track</b>       | Sejong Science High School                                 |

## Work Experience

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025.08 - Present | <b>Planning &amp; Scheduling Solution Consultant</b><br>VMS Solutions Inc.   Solution Division   Consultant Part<br>Ministry of National Defense   Tech. Research Personnel [2025.08 - 2026.02]<br>System designer & PMO: digital twin platform, production-logistics integrated simulator, simulation-based AI scheduler, SaaS product<br>– SK Hynix Digital Twin Platform: production-logistics simulator PM, use-case discovery, KPI definition, data consistency metrics, MES legacy data analysis & simulator integration, AI prediction model data specification<br>– Production-logistics integrated simulation design<br>– Production simulation engine architecture redesign & development participation<br>– Planned and authored 2 government R&D proposals on Agentic AI (manufacturing-specific Agentic AI, LLM-based production plan explanation module / with University of Toronto)<br>– AI extensibility review & design for simulation-based APS, new engine feature definition, participation in GenAI-based development<br>– LLM-based schedule analysis & report generation system performance review and feedback |
| 2025.02 - 2025.07 | <b>Planning &amp; Scheduling Solution Consultant</b><br>VMS Solutions Inc.   Solution Division   Micron Part<br>Ministry of National Defense   Tech. Research Personnel [2025.02 - 2025.07]<br>Solution consultant: simulation-based real-time APS maintenance, system configuration remote support<br>– Micron global fab simulation-based real-time APS (advanced production and scheduling) system logic maintenance and remote system configuration support                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

- 2021.03 - 2025.02 **Researcher**  
**KAIST | MSS (Manufacturing Service Systems) Lab.**  
 Ministry of National Defense | Tech. Research Personnel [2023.03 - 2025.02]  
 Lead researcher: AI-based dynamic schedulers, optimization-based heuristics, autonomous manufacturing scheduling systems  
 – Led full project lifecycle (planning-to-validation): 9 industry-academia projects (planned 5, PM 4), 4 government projects (planned 3, PM 4)  
 – Fundamental research on Neural Combinatorial Optimization based on GNN and RL  
 – Designed general-purpose scheduling agent and Meta-RL methodology for flexible adaptation to manufacturing environment changes  
 – Designed AI-based demand forecasting, production planning, and job scheduling optimization algorithms; validated business impact using real production data (Samsung Electronics, LG Electronics, Samsung Heavy Industries, etc.)  
 – Provided technical review supporting client executives' strategic decisions and delivered system design proposals  
 – Introduced structured project management processes (OneDrive & Notion-based) to the lab for the first time  
 – First Ph.D. graduate of the lab; led project planning, proposal writing, and research beyond typical researcher scope  
 – Won 10-year NRF 'Hanwoomul' long-term project after analyzing rejection feedback and reinforcing proposal logic  
 – 2 first-author IEEE journal papers, 7 international conference presentations
- 2016.04 - 2021.02 **Researcher**  
**SKKU | SCO (Systems Control & Optimization) Lab.**  
 Scheduling algorithm developer: meta-heuristics, optimization; System designer: digital twin for manufacturing, dynamic rescheduling systems  
 – 5 industry-academia projects (planned 2, PM 2), 4 government projects (planned 2, PM 1)  
 – Led foundational design and operations optimization research for 3D printer-based smart factory digital twin systems  
 – Developed rescheduling algorithms for handling disruptions (machine failures, order cancellations) in flexible manufacturing systems (FMS)  
 – Translated physical manufacturing constraints into mathematical models and validated in virtual environments  
 – Defined virtual factory operation scenarios and conducted simulation-based operational efficiency reviews  
 – Designed and verified real-time scheduling methodologies using genetic algorithms and heuristics

## Projects

- 2025.06 - 2025.12 **Digital twin platform construction: production-logistics simulation-based operation twin for semiconductor fab**  
 7m  
 VMS Solutions Inc. (with SK Hynix, SKT, SK AX, Calro)
- 2025.03 - **Software product design/development: simulation-based scheduling for semiconductor manufacturing**  
 VMS Solutions Inc.
- 2025.02 - 2025.06 **Simulation-based real-time scheduler for semiconductor fab**  
 VMS Solutions Inc. (with Micron)  
 5m (5y)
- 2024.05 - 2025.02 **Autonomous scheduling for swift and efficient adaptation to dynamic manufacturing environments**  
 KAIST (with NRF of Korea)  
 10m (10y)
- 2024.03 - 2025.02 **AI-based algorithm for optimal operation plans for various scenarios**  
 KAIST (with Samsung Electronics)  
 1y
- 2023.07 - 2024.06 **Reinforcement learning for unrelated parallel machine scheduling problems with sequence-dependent setup times and machine eligibility**  
 KAIST (with VMS Solutions)  
 1y
- 2023.07 - 2024.02 **Production planning with AI**  
 KAIST (with LG Electronics)  
 8m
- 2022.05 - 2023.02 **Graph-based reinforcement learning algorithm for real-time job shop scheduling**  
 KAIST (with NRF of Korea)  
 10m
- 2022.04 - 2023.03 **Reinforcement learning for job shop scheduling**  
 KAIST (with VMS Solutions)  
 1y
- 2022.03 - 2022.03 **Development of a wiring optimization algorithm for X-DEC slim layout**  
 KAIST (with SK Hynix)  
 1m (7m)
- 2022.03 - 2025.02 **Reinforcement learning-based meta-scheduling for manufacturing systems**  
 KAIST (with NRF of Korea)  
 3y

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|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022.03 -<br>2023.07<br>1y5m (2y) | <b>Reinforcement learning for project scheduling</b><br>KAIST (with Samsung Heavy Industries)                                                     |
| 2021.07 -<br>2021.11<br>5m        | <b>Development of a reinforcement learning algorithm for workload balancing of ship cargo production</b><br>KAIST (with Samsung Heavy Industries) |
| 2020.06 -<br>2021.02<br>9m        | <b>Optimal machine assignment with machine learning algorithms</b><br>KAIST (with VMS Solutions)                                                  |
| 2019.06 -<br>2022.05<br>3y        | <b>Cyber-physical assembly and logistics systems in global supply chains</b><br>KAIST (with MOTIE of Korea, Yura)                                 |
| 2019.09 -<br>2020.02<br>6m        | <b>Optimal weight sets for dispatching rules with multiple KPIs</b><br>KAIST (with VMS Solutions)                                                 |
| 2019.04 -<br>2019.12<br>9m (1y9m) | <b>Big data-based simulation and optimization technology for smart manufacturing</b><br>SKKU (with MOTIE of Korea, Samsung SDI)                   |
| 2019.03 -<br>2022.02<br>3y        | <b>Development of scheduling theory and algorithms with reinforcement learning for manufacturing systems</b><br>SKKU (with NRF of Korea)          |
| 2018.07 -<br>2018.09<br>3m        | <b>Methodology for dispatching rules' weights</b><br>SKKU (with SK Hynix)                                                                         |
| 2018.07 -<br>2019.02<br>8m        | <b>Framework development for KPI analysis with various weights on dispatching rules</b><br>SKKU (with VMS Solutions)                              |
| 2017.07 -<br>2018.02<br>8m        | <b>Analysis of KPIs according to weights on dispatching rules for LCD manufacturing</b><br>SKKU (with VMS Solutions)                              |
| 2017.07 -<br>2018.01<br>7m        | <b>Design and analysis for operations optimizations of smart factory testbed</b><br>SKKU (with MSIP of Korea)                                     |
| 2016.11 -<br>2019.10<br>3y        | <b>Development of scheduling and rescheduling algorithms for 3D printer-based smart factory</b><br>SKKU (with NRF of Korea)                       |
| 2016.07 -<br>2017.02<br>8m        | <b>Development of algorithms for detecting and improving inefficient schedules in LCD processes</b><br>SKKU (with VMS Solutions)                  |
| 2016.04 -<br>2018.05<br>2y2m (3y) | <b>Development of open FaaS IoT service platform for mass personalization</b><br>SKKU (with MSIP of Korea)                                        |

## Academic Works

|      |                                                                                                                                              |                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 2024 | <b>Graph-based imitation learning for real-time job shop dispatcher</b><br>IEEE Transactions on Automation Science and Engineering           | 1st Author<br><a href="#">[LINK]</a> |
| 2024 | <b>Tree-based Dispatcher for Job Shop Scheduling</b><br>IEEE International Conference on Automation Science and Engineering (CASE)           | 1st Author                           |
| 2024 | <b>Tree-based dispatcher for solving job shop scheduling problems</b><br>the Spring Conference of Korean Institute of Industrial Engineers   | 1st Author                           |
| 2023 | <b>Active schedule-based imitation learning for job shop scheduling</b><br>the Spring Conference of Korean Institute of Industrial Engineers | 1st Author                           |
| 2022 | <b>Imitation learning for real-time job shop scheduling using graph-based representation</b><br>Winter Simulation Conference                 | 1st Author<br><a href="#">[LINK]</a> |
| 2022 | <b>Job shop scheduling using graph-based imitation learning</b><br>INFORMS Annual Meeting                                                    | 1st Author                           |

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 2022 | <b>A multi-manned assembly line worker assignment and balancing problem with positional constraints</b><br>IEEE Robotics and Automation Letters                                  | Co-Author<br><a href="#">[LINK]</a>  |
| 2022 | <b>Reinforcement learning for resource leveling in multiple projects</b><br>the Spring Conference of Korean Institute of Industrial Engineers                                    | 1st Author                           |
| 2022 | <b>Dynamic job shop scheduling using graph-based imitation learning</b><br>the Spring Conference of Korean Institute of Industrial Engineers                                     | 1st Author                           |
| 2021 | <b>Machine learning-based periodic setup changes for semiconductor manufacturing machines</b><br>Winter Simulation Conference                                                    | 1st Author<br><a href="#">[LINK]</a> |
| 2021 | <b>Resource leveling in shipyard cargo hold process through reinforcement learning</b><br>the Autumn Conference of Korean Institute of Industrial Engineers                      | Co-Author                            |
| 2021 | <b>Assembly line worker assignment and balancing problem with positional constraints</b><br>Advances in Production Management Systems (APMS)                                     | Co-Author<br><a href="#">[LINK]</a>  |
| 2021 | <b>Operation and optimization of the automotive parts assembly line considering worker skill levels</b><br>the Summer Conference of Korea CDE                                    | Co-Author                            |
| 2020 | <b>A sequential search method of dispatching rules for scheduling of LCD manufacturing systems</b><br>IEEE Transactions on Semiconductor Manufacturing                           | 1st Author<br><a href="#">[LINK]</a> |
| 2020 | <b>A simulation-based sequential search method for multi-objective scheduling problems of manufacturing systems</b><br>Winter Simulation Conference                              | 1st Author                           |
| 2020 | <b>Workforce assignment for automotive parts assembly lines</b><br>the Winter Conference of Korea CDE                                                                            | Co-Author                            |
| 2020 | <b>Digital twin-based cyber physical production system architectural framework for personalized production</b><br>The International Journal of Advanced Manufacturing Technology | Co-Author<br><a href="#">[LINK]</a>  |
| 2020 | <b>Workforce assignment with a different skill level for automotive parts assembly lines</b><br>Advances in Production Management Systems (APMS)                                 | Co-Author<br><a href="#">[LINK]</a>  |
| 2019 | <b>A sequential search framework for selecting weights of dispatching rules in manufacturing systems</b><br>Winter Simulation Conference                                         | 1st Author<br><a href="#">[LINK]</a> |
| 2019 | <b>A genetic algorithm for hybrid flow shop scheduling with multiple assembly stages</b><br>the Autumn Conference of Korean Institute of Industrial Engineers                    | 1st Author                           |
| 2018 | <b>Vulnerability analysis of evacuation transportation networks</b><br>International Journal of Industrial Engineering-Theory Applications and Practice                          | Co-Author                            |
| 2018 | <b>A framework for performance analysis of dispatching rules in manufacturing systems</b><br>Winter Simulation Conference                                                        | 1st Author                           |
| 2018 | <b>Rescheduling algorithms for 3D printer-based manufacturing systems</b><br>the Summer Conference of Korea CDE                                                                  | 1st Author                           |
| 2018 | <b>Scheduling algorithms for 3D printer-based manufacturing systems</b><br>the Spring Conference of Korean Institute of Industrial Engineers                                     | Co-Author                            |
| 2017 | <b>Rescheduling of flexible flow shop with sequence-dependent setup times and job splitting</b><br>Winter Simulation Conference                                                  | Co-Author<br><a href="#">[LINK]</a>  |
| 2017 | <b>3D printer based assembly process scheduling algorithm development</b><br>the Winter Conference of Korea CDE                                                                  | Co-Author                            |

## Honors & Awards

|         |                                                                                                          |                                                 |
|---------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 2025.11 | <b>Second Prize, Ph.D. Thesis Competition</b> , Learning Schedulers for Job Shop Scheduling Problems     | Korean Institute of Industrial Engineers (KIIE) |
| 2023.12 | <b>First Prize, 2023 Simulation Challenge</b> , Q-time-based Gating Control in Semiconductor Fabrication | Winter Simulation Conference (WSC), Micron      |

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|-------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 2023.09           | <b>Third Prize, 2023 AI Competition: Solving Real-world Problems</b> , Integrated Planning Module using AI        | Hankook & Company                           |
| 2022.06 - 2023.02 | <b>Ph.D. Candidate Research Incentive Support</b>                                                                 | National Research Foundation (NRF) of Korea |
| 2022.09           | <b>Second Prize, 2022 Poster Competition: Industry/Social Problems</b> , RL for Resource Leveling in Shipbuilding | KAIST                                       |
| 2022.05           | <b>Certificate of Appreciation: Successful Project</b> , Workload Balancing of Ship Cargo Production              | Samsung Heavy Industries                    |
| 2021.03 - 2025.02 | <b>Full-tuition Entrance Scholarship</b> , Merit-based Entrance Scholarship (Ph.D.)                               | KAIST                                       |
| 2019.03, 2019.09  | <b>Graduate Assistant Tuition Waiver</b> , Tuition Waiver (M.S. & Ph.D Integrated Program)                        | SKKU                                        |
| 2018.03, 2019.03  | <b>Simsan Scholarship</b> , Scholarship (M.S. & Ph.D Integrated Program)                                          | SKKU                                        |
| 2018.03           | <b>Systems &amp; Management Engineering Scholarship</b> , Scholarship (M.S. & Ph.D Integrated Program)            | SKKU                                        |
| 2017.03           | <b>Industrial Engineering Alumni Association Scholarship</b> , Scholarship (M.S. & Ph.D Integrated Program)       | SKKU                                        |
| 2016.07           | <b>Third Prize, 2015-2016 PACE RSMS Competition (Second Year): Customer Insight</b>                               | General Motors (GM)                         |
| 2016.07           | <b>Third Prize, 2015-2016 PACE RSMS Competition (Second Year): Manufacturing Engineering</b>                      | General Motors (GM)                         |
| 2013.03 - 2016.08 | <b>Full-tuition Entrance Scholarship</b> , Merit-based Entrance Scholarship (B.S.)                                | SKKU                                        |

## Activities & Leadership

|                   |                                  |                                                                            |
|-------------------|----------------------------------|----------------------------------------------------------------------------|
| 2022.03 - 2023.08 | <b>Researcher Representative</b> | Manufacturing and Service Systems Lab. (KAIST)                             |
| 2021.10 – 2021.12 | <b>Technical Mentor</b>          | Public Data Internship Program (Korea National Information Society Agency) |
| 2021.03 – 2024.12 | <b>Teaching Assistant (TA)</b>   | Scheduling (KAIST)                                                         |
| 2018.09 – 2019.06 | <b>Teaching Assistant (TA)</b>   | Supply Chain Management (SKKU)                                             |
| 2018.03 – 2019.02 | <b>President</b>                 | Industrial Engineering Graduate Student Council (SKKU)                     |
| 2018.03 – 2018.06 | <b>Teaching Assistant (TA)</b>   | Operations Management (SKKU)                                               |
| 2016.10 – 2018.12 | <b>Teaching Assistant (TA)</b>   | Gender Awareness Education (SKKU)                                          |
| 2017.09 – 2017.12 | <b>Teaching Assistant (TA)</b>   | Engineering Economy (SKKU)                                                 |
| 2017.03 – 2017.06 | <b>Teaching Assistant (TA)</b>   | Operations Research and Practice 1 (SKKU)                                  |
| 2016.09 – 2016.12 | <b>Teaching Assistant (TA)</b>   | Engineering Economy (SKKU)                                                 |
| 2015.03 – 2016.08 | <b>Working Group Member</b>      | Systems Management Engineering Student Council (SKKU)                      |
| 2015.03 – 2016.08 | <b>President</b>                 | Turbo: Basketball Club (SKKU)                                              |
| 2014.08 – 2020.02 | <b>Working Group Member</b>      | Turbo: Basketball Club (SKKU)                                              |
| 2013.03 – 2015.02 | <b>Working Group Member</b>      | College of Engineering Student Council (SKKU)                              |
| 2012.03 - 2013.02 | <b>President</b>                 | Student Council (High School, 11th Grade)                                  |
| 2010.03 - 2011.02 | <b>Vice President</b>            | Student Council (Middle School, 9th Grade)                                 |

## Skills

|                                   |                                                            |
|-----------------------------------|------------------------------------------------------------|
| <b>Programming</b>                | Python, C#, SQL, LaTeX, R                                  |
| <b>Frameworks &amp; Libraries</b> | PyTorch, Matplotlib, Pandas, NumPy, Gurobi, CPLEX          |
| <b>Simulation &amp; Tools</b>     | VMS Mozart (APS), Plant Sim, AutoMod, MS Office, Photoshop |
| <b>DevOps &amp; Productivity</b>  | Git / GitHub, Docker, Claude Code, n8n, Notion             |
| <b>Languages</b>                  | Korean (Native), English (Professional)                    |